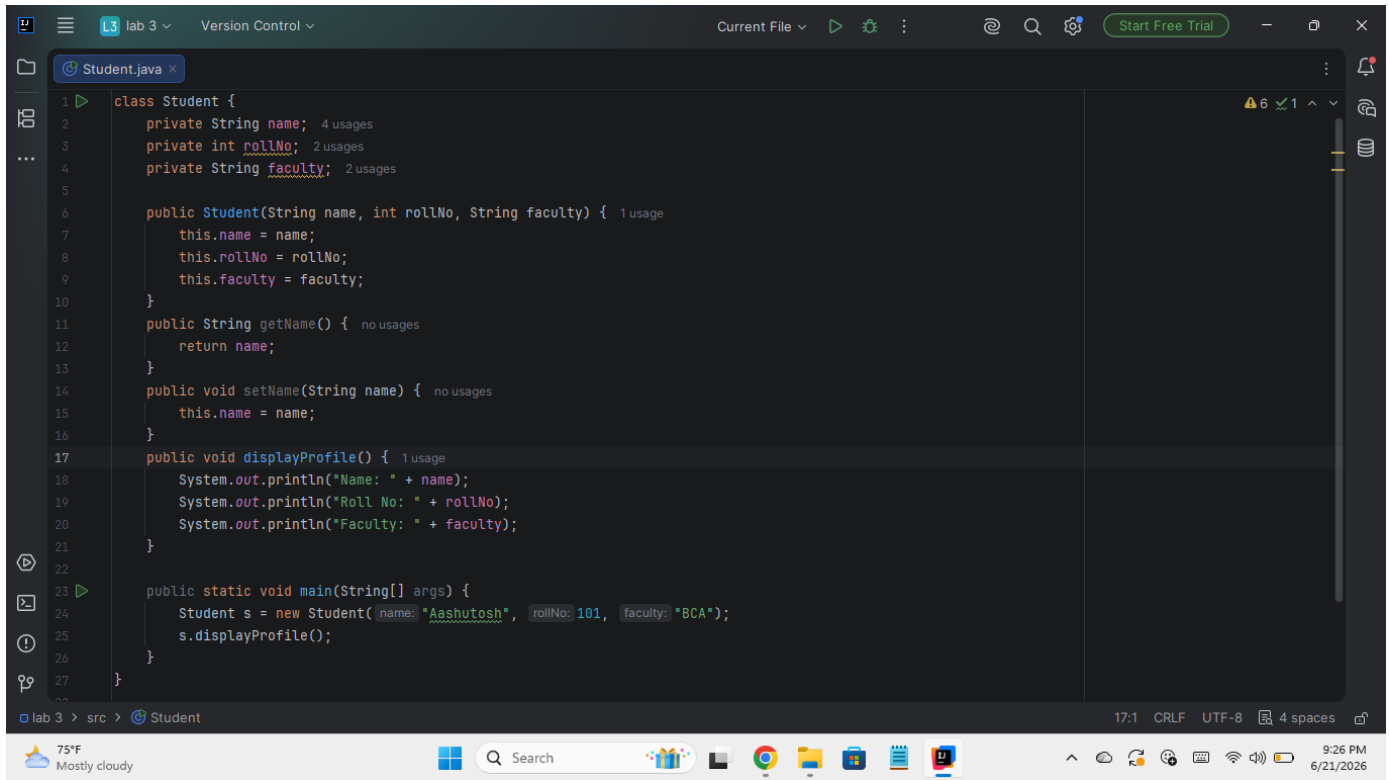


JAVA

1.Student profile

Program:

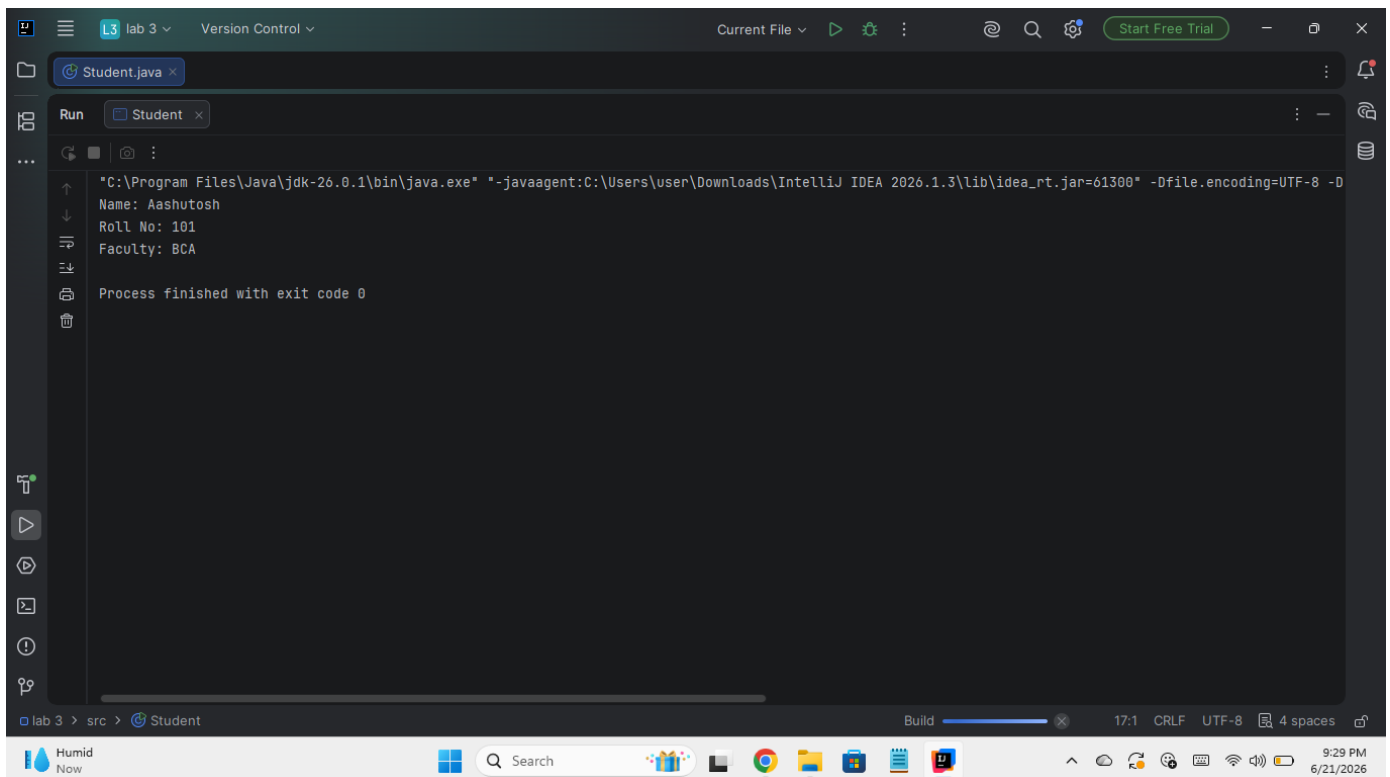


```
1 class Student {
2     private String name; 4 usages
3     private int rollNo; 2 usages
4     private String faculty; 2 usages
5
6     public Student(String name, int rollNo, String faculty) { 1 usage
7         this.name = name;
8         this.rollNo = rollNo;
9         this.faculty = faculty;
10    }
11    public String getName() { no usages
12        return name;
13    }
14    public void setName(String name) { no usages
15        this.name = name;
16    }
17    public void displayProfile() { 1 usage
18        System.out.println("Name: " + name);
19        System.out.println("Roll No: " + rollNo);
20        System.out.println("Faculty: " + faculty);
21    }
22
23    public static void main(String[] args) {
24        Student s = new Student("Aashutosh", rollNo: 101, faculty: "BCA");
25        s.displayProfile();
26    }
27 }
```

lab 3 > src > Student 17:1 CRLF UTF-8 4 spaces

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Output:



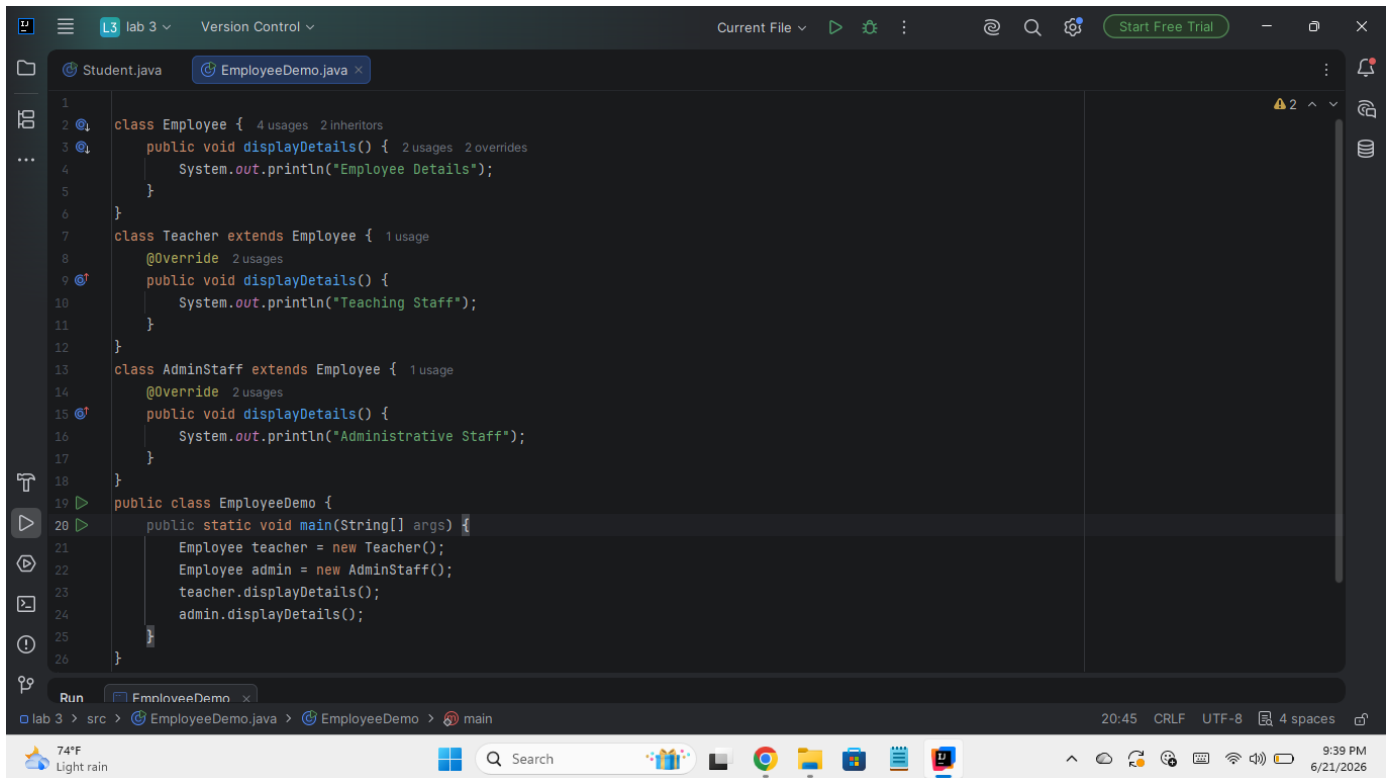
```
Run Student
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\Downloads\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=61300" -Dfile.encoding=UTF-8 -D
Name: Aashutosh
Roll No: 101
Faculty: BCA
Process finished with exit code 0
```

lab 3 > src > Student Build 17:1 CRLF UTF-8 4 spaces

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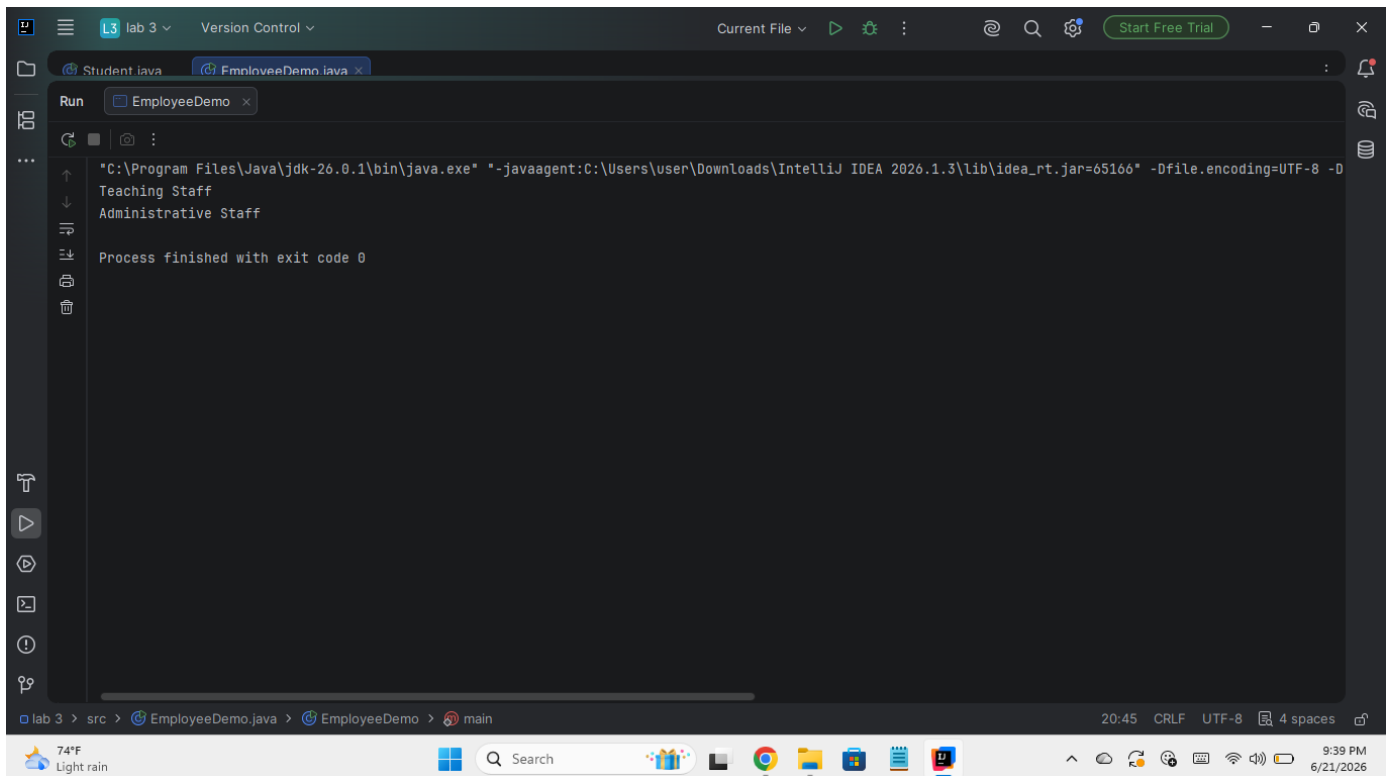
2.Employee id

Program:



```
1 class Employee { 4 usages 2 inheritors
2     public void displayDetails() { 2 usages 2 overrides
3         System.out.println("Employee Details");
4     }
5 }
6
7 class Teacher extends Employee { 1 usage
8     @Override 2 usages
9     public void displayDetails() {
10         System.out.println("Teaching Staff");
11     }
12 }
13
14 class AdminStaff extends Employee { 1 usage
15     @Override 2 usages
16     public void displayDetails() {
17         System.out.println("Administrative Staff");
18     }
19 }
20
21 public class EmployeeDemo {
22     public static void main(String[] args) {
23         Employee teacher = new Teacher();
24         Employee admin = new AdminStaff();
25         teacher.displayDetails();
26         admin.displayDetails();
27     }
28 }
```

Output:



```
Run EmployeeDemo
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\Downloads\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=65166" -Dfile.encoding=UTF-8 -D
Teaching Staff
Administrative Staff
Process finished with exit code 0
```

3.Students courses

Program:

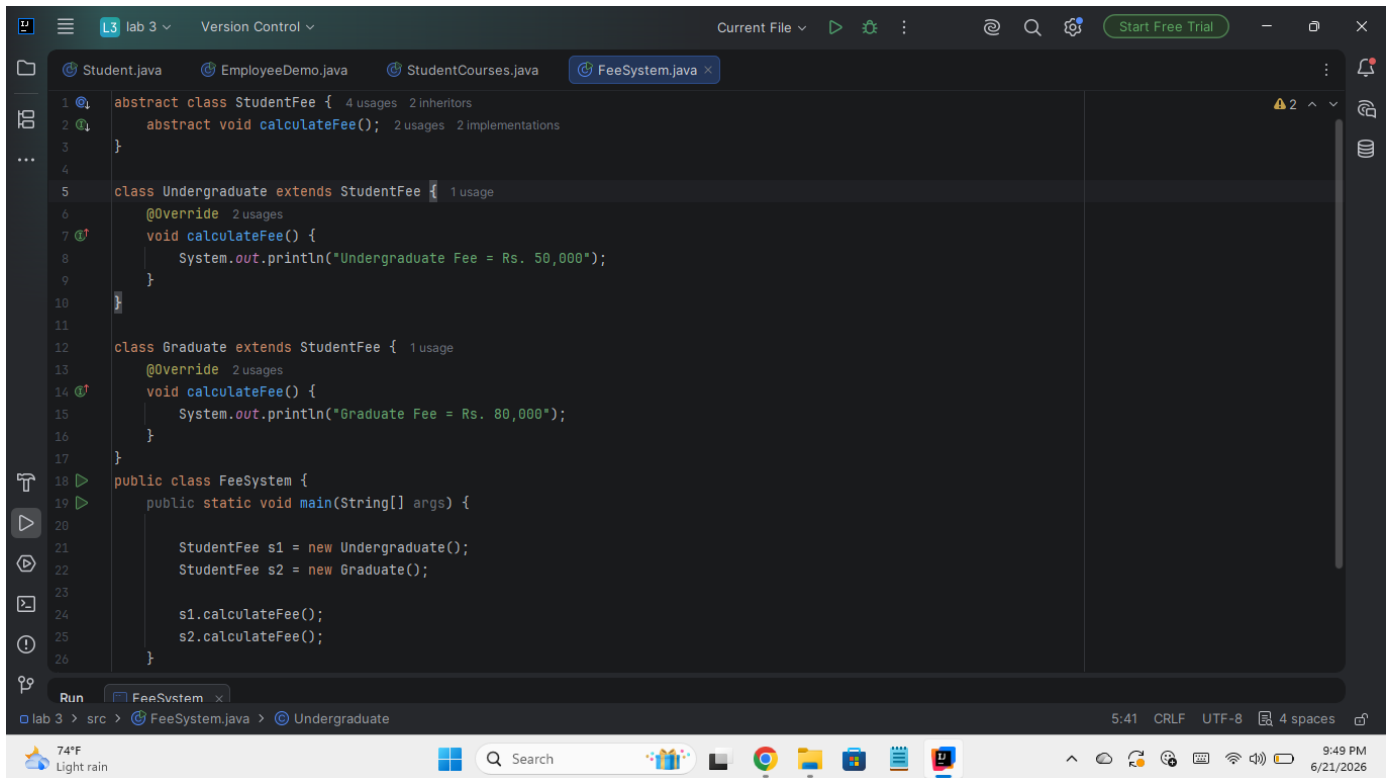
```
1 import java.util.ArrayList;
2
3 class Course {
4     String courseName;
5     Course(String courseName) {
6         this.courseName = courseName;
7     }
8 }
9
10 class StudentData {
11     String name;
12     ArrayList<Course> courses = new ArrayList<>();
13     StudentData(String name) {
14         this.name = name;
15     }
16     void addCourse(Course course) {
17         courses.add(course);
18     }
19     void displayCourses() {
20         System.out.println("Student Name: " + name);
21         System.out.println("Registered Courses:");
22         for (Course c : courses) {
23             System.out.println("- " + c.courseName);
24         }
25     }
26 }
27
28 public class StudentCourses {
29     public static void main(String[] args) {
30         StudentData student = new StudentData("Aashutosh");
31         student.addCourse(new Course("Java"));
32         student.addCourse(new Course("DBMS"));
33         student.addCourse(new Course("Networking"));
34         student.displayCourses();
35     }
36 }
```

Output:

```
Run StudentCourses
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\Downloads\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=53593" -Dfile.encoding=UTF-8 -D
Student Name: Aashutosh
Registered Courses:
- Java
- DBMS
- Networking
Process finished with exit code 0
```

4. Fee Structure

program:



```
1 abstract class StudentFee { 4 usages 2 inheritors
2     abstract void calculateFee(); 2 usages 2 implementations
3 }
4
5 class Undergraduate extends StudentFee { 1 usage
6     @Override 2 usages
7     void calculateFee() {
8         System.out.println("Undergraduate Fee = Rs. 50,000");
9     }
10 }
11
12 class Graduate extends StudentFee { 1 usage
13     @Override 2 usages
14     void calculateFee() {
15         System.out.println("Graduate Fee = Rs. 80,000");
16     }
17 }
18
19 public class FeeSystem {
20     public static void main(String[] args) {
21
22         StudentFee s1 = new Undergraduate();
23         StudentFee s2 = new Graduate();
24
25         s1.calculateFee();
26         s2.calculateFee();
27     }
28 }
```

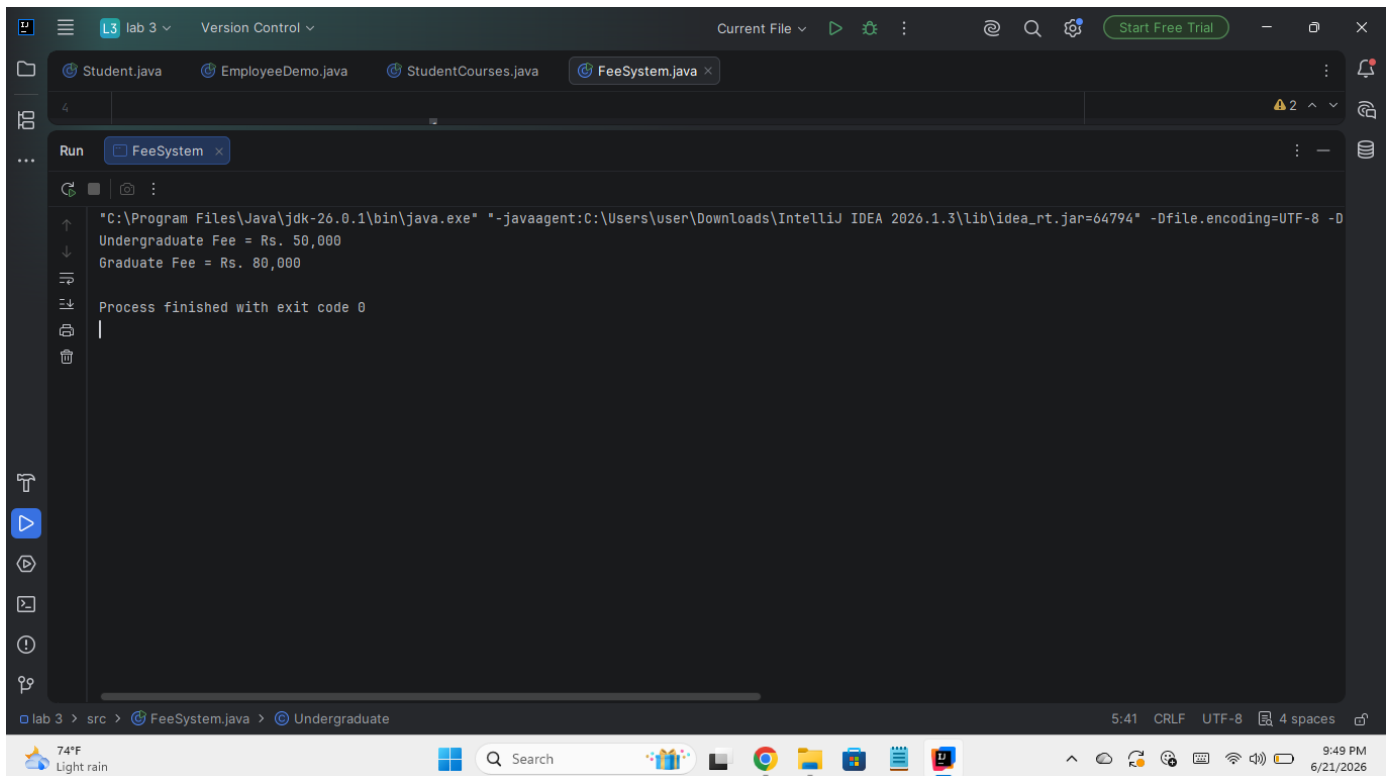
Run FeeSystem x

lab 3 > src > FeeSystem.java > Undergraduate

5:41 CRLF UTF-8 4 spaces

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Output:



```
Run FeeSystem x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\Downloads\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=64794" -Dfile.encoding=UTF-8 -D
Undergraduate Fee = Rs. 50,000
Graduate Fee = Rs. 80,000
Process finished with exit code 0
```

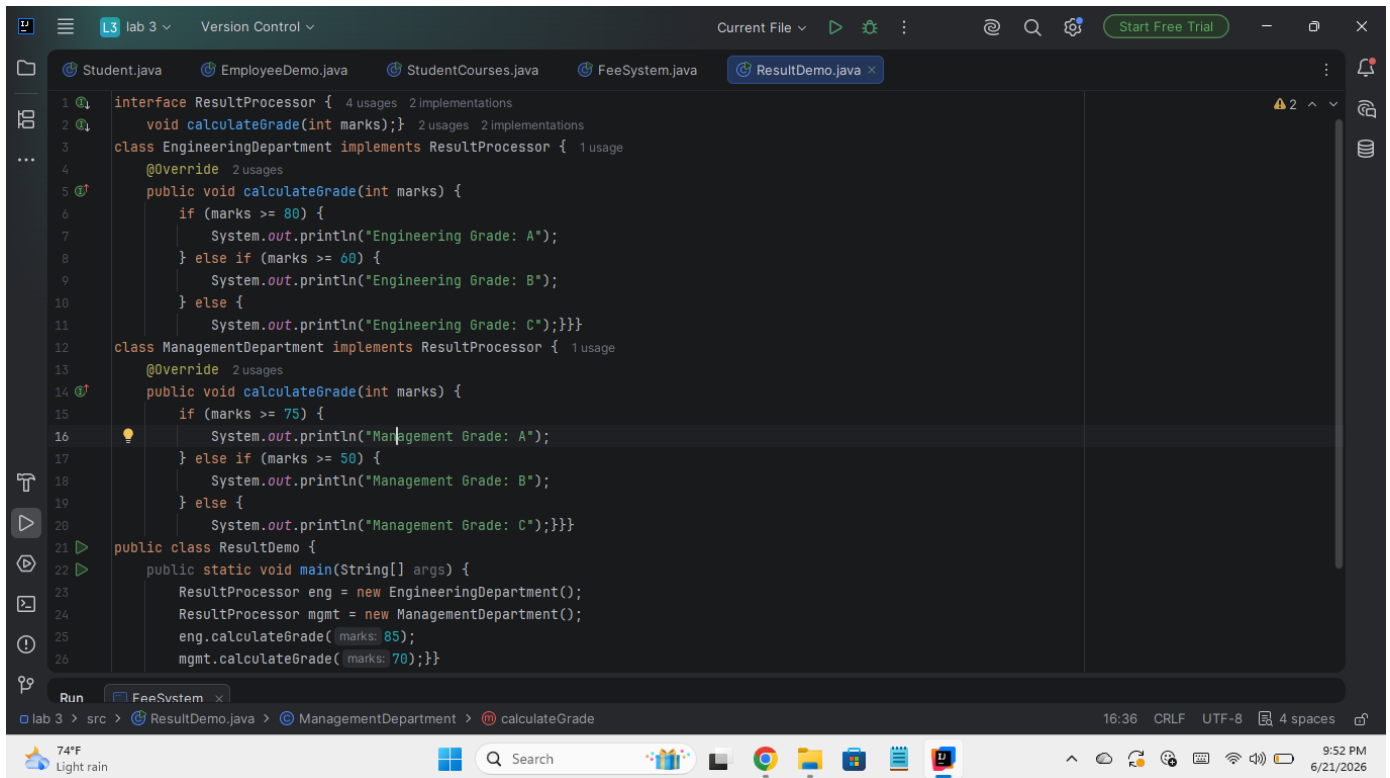
lab 3 > src > FeeSystem.java > Undergraduate

5:41 CRLF UTF-8 4 spaces

74°F Light rain 9:49 PM 6/21/2026

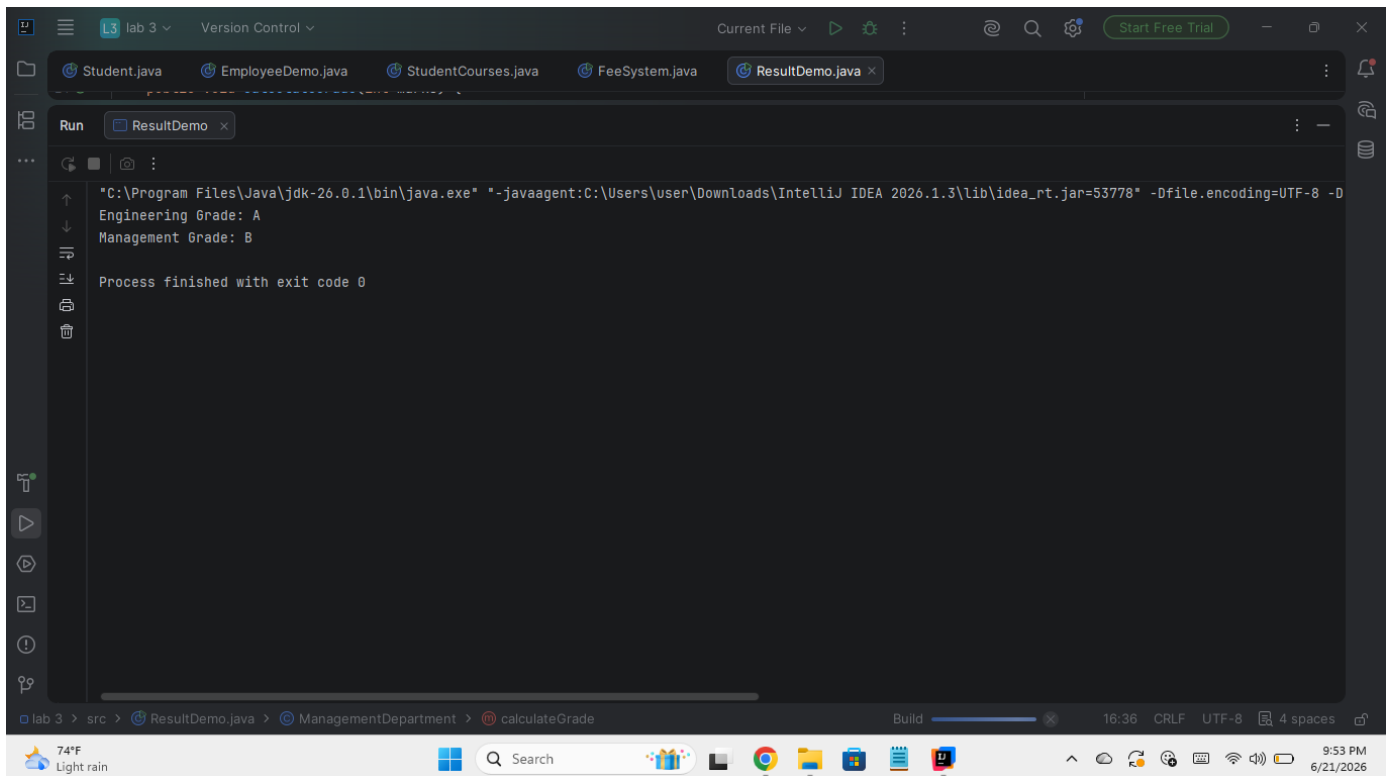
5. Different department grade

Program:



```
1 interface ResultProcessor { 4 usages 2 implementations
2     void calculateGrade(int marks);} 2 usages 2 implementations
3
4 class EngineeringDepartment implements ResultProcessor { 1 usage
5     @Override 2 usages
6     public void calculateGrade(int marks) {
7         if (marks >= 80) {
8             System.out.println("Engineering Grade: A");
9         } else if (marks >= 60) {
10            System.out.println("Engineering Grade: B");
11        } else {
12            System.out.println("Engineering Grade: C");}}}
13
14 class ManagementDepartment implements ResultProcessor { 1 usage
15     @Override 2 usages
16     public void calculateGrade(int marks) {
17         if (marks >= 75) {
18             System.out.println("Management Grade: A");
19         } else if (marks >= 50) {
20             System.out.println("Management Grade: B");
21         } else {
22             System.out.println("Management Grade: C");}}}
23
24 public class ResultDemo {
25     public static void main(String[] args) {
26         ResultProcessor eng = new EngineeringDepartment();
27         ResultProcessor mgmt = new ManagementDepartment();
28         eng.calculateGrade( marks: 85);
29         mgmt.calculateGrade( marks: 70);}}
```

Output:

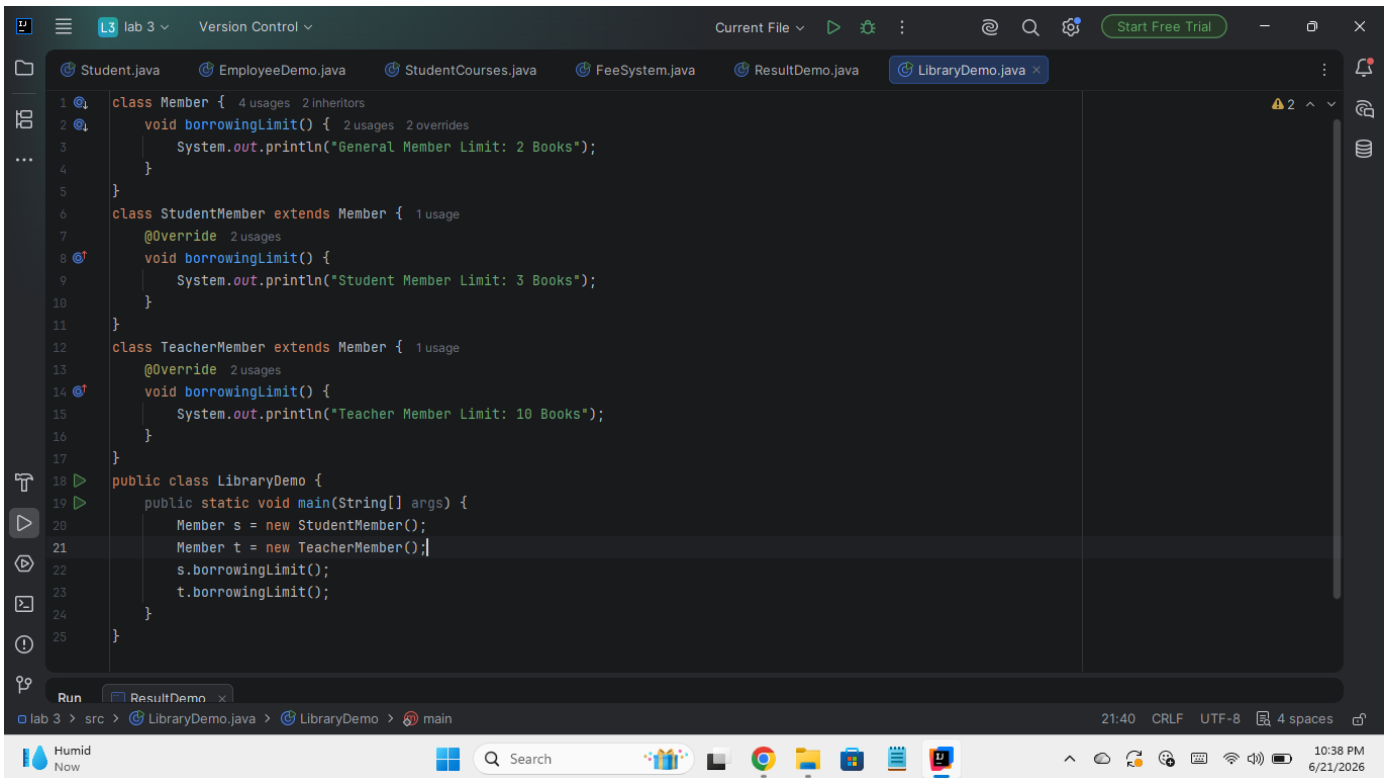


```
Run ResultDemo x
"\"C:\\Program Files\\Java\\jdk-26.0.1\\bin\\java.exe\" \"-javaagent:C:\\Users\\user\\Downloads\\IntelliJ IDEA 2026.1.3\\lib\\idea_rt.jar=53778\" -Dfile.encoding=UTF-8 -D
Engineering Grade: A
Management Grade: B

Process finished with exit code 0
```

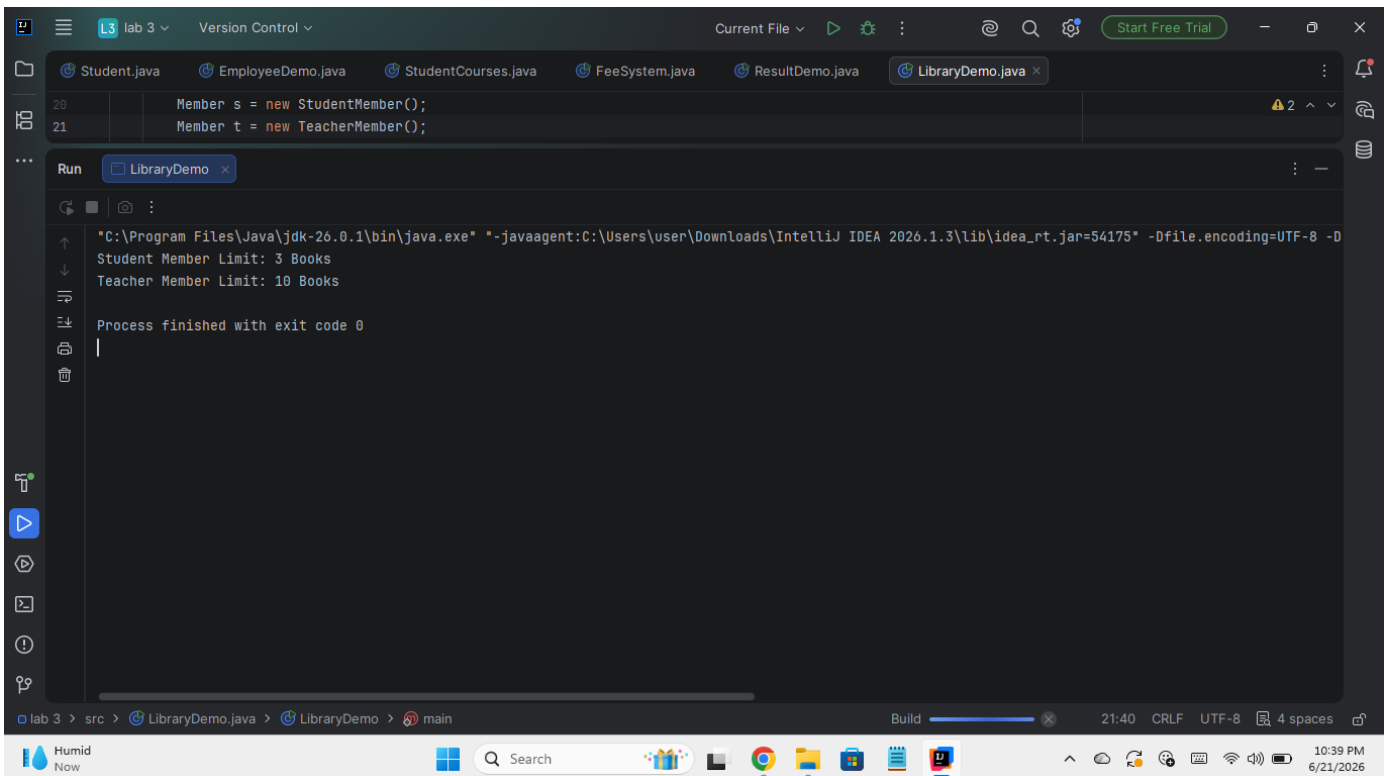
6 Library members

Program:



```
1 class Member { 4 usages 2 inheritors
2 void borrowingLimit() { 2 usages 2 overrides
3     System.out.println("General Member Limit: 2 Books");
4 }
5 }
6 class StudentMember extends Member { 1 usage
7     @Override 2 usages
8     void borrowingLimit() {
9         System.out.println("Student Member Limit: 3 Books");
10    }
11 }
12 class TeacherMember extends Member { 1 usage
13     @Override 2 usages
14     void borrowingLimit() {
15         System.out.println("Teacher Member Limit: 10 Books");
16    }
17 }
18 public class LibraryDemo {
19     public static void main(String[] args) {
20         Member s = new StudentMember();
21         Member t = new TeacherMember();
22         s.borrowingLimit();
23         t.borrowingLimit();
24     }
25 }
```

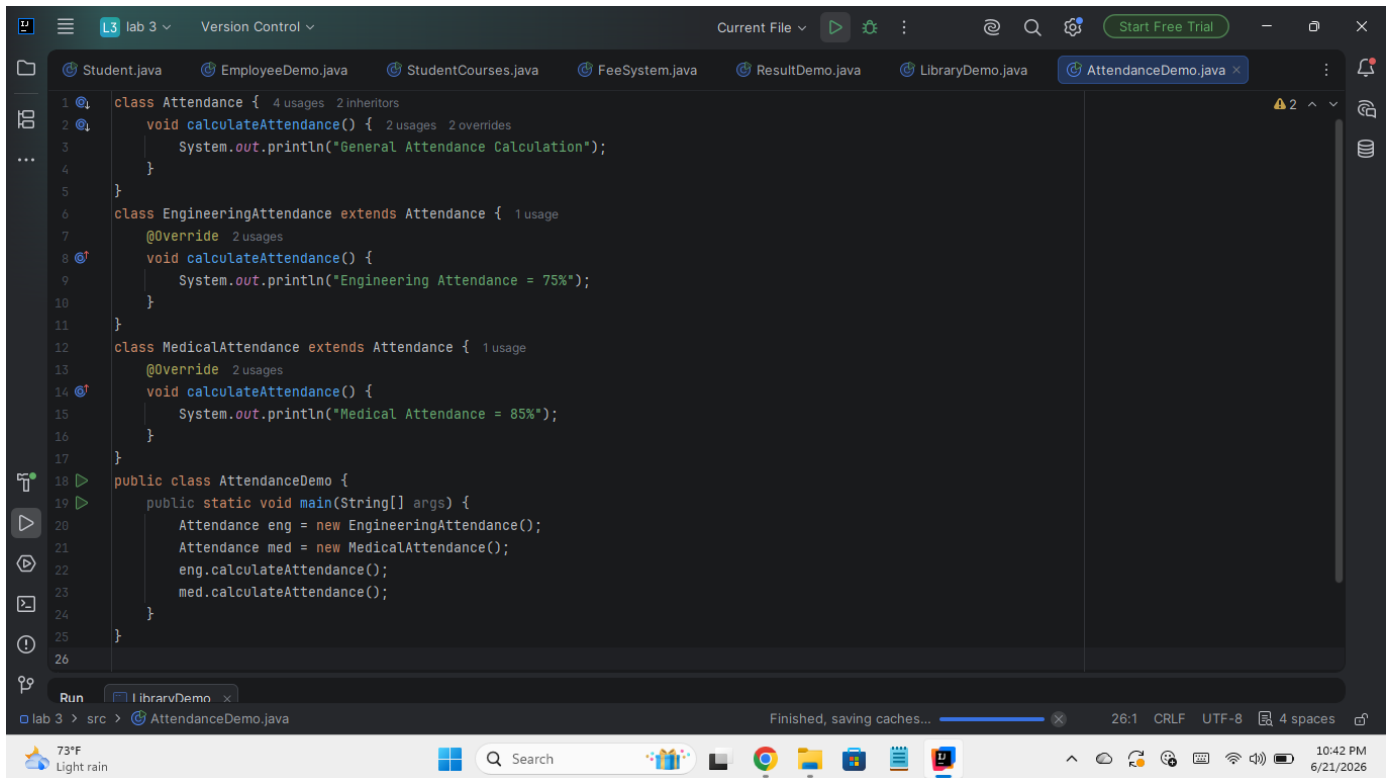
Output:



```
Run LibraryDemo x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\Downloads\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=54175" -Dfile.encoding=UTF-8 -D
Student Member Limit: 3 Books
Teacher Member Limit: 10 Books
Process finished with exit code 0
```

7 Attendance percentage

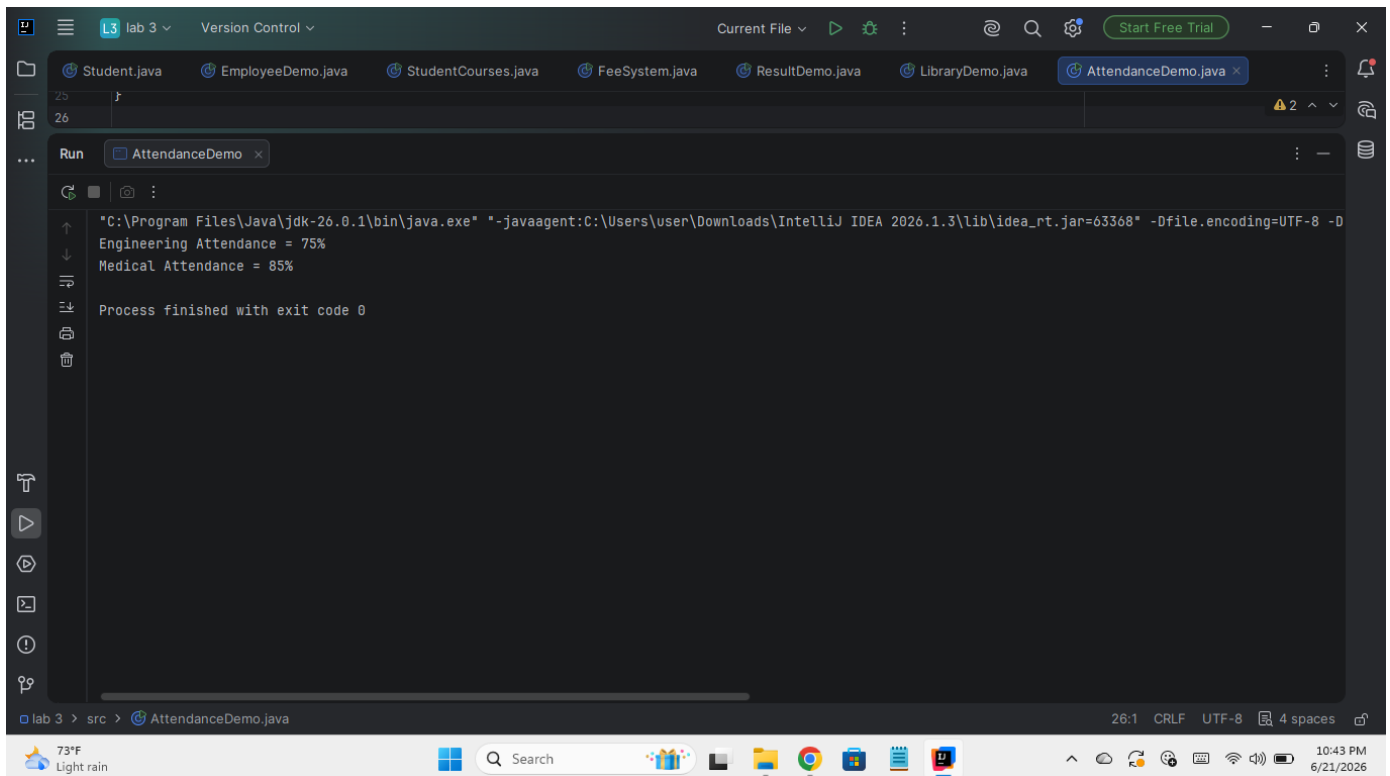
Program:



```
1 class Attendance { 4 usages 2 inheritors
2     void calculateAttendance() { 2 usages 2 overrides
3         System.out.println("General Attendance Calculation");
4     }
5 }
6 class EngineeringAttendance extends Attendance { 1 usage
7     @Override 2 usages
8     void calculateAttendance() {
9         System.out.println("Engineering Attendance = 75%");
10    }
11 }
12 class MedicalAttendance extends Attendance { 1 usage
13     @Override 2 usages
14     void calculateAttendance() {
15         System.out.println("Medical Attendance = 85%");
16    }
17 }
18 public class AttendanceDemo {
19     public static void main(String[] args) {
20         Attendance eng = new EngineeringAttendance();
21         Attendance med = new MedicalAttendance();
22         eng.calculateAttendance();
23         med.calculateAttendance();
24     }
25 }
26
```

Run librayDemo... Finished, saving caches... 26:1 CRLF UTF-8 4 spaces

Output:

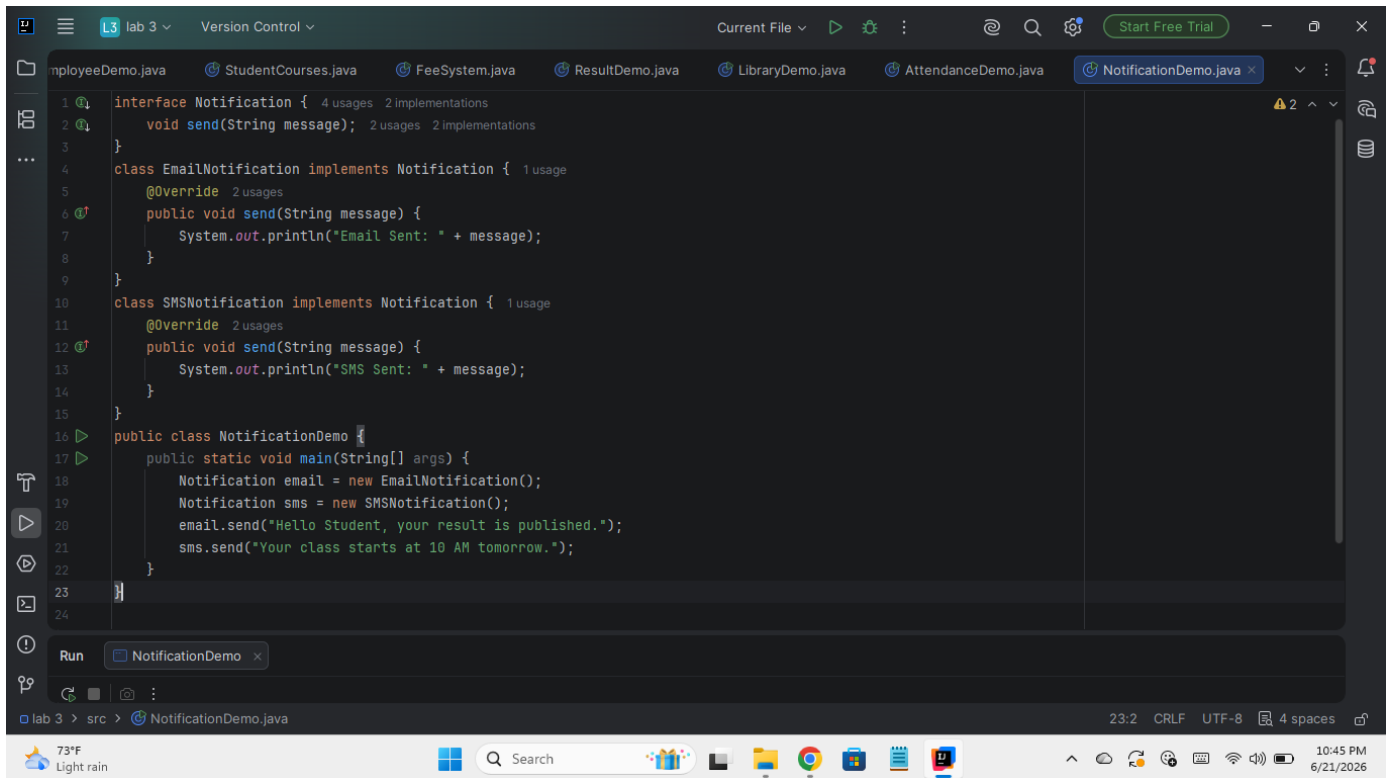


```
Run AttendanceDemo
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\Downloads\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=63368" -Dfile.encoding=UTF-8 -D
Engineering Attendance = 75%
Medical Attendance = 85%
Process finished with exit code 0
```

lab 3 > src > AttendanceDemo.java 26:1 CRLF UTF-8 4 spaces

8 Notifications

Program:

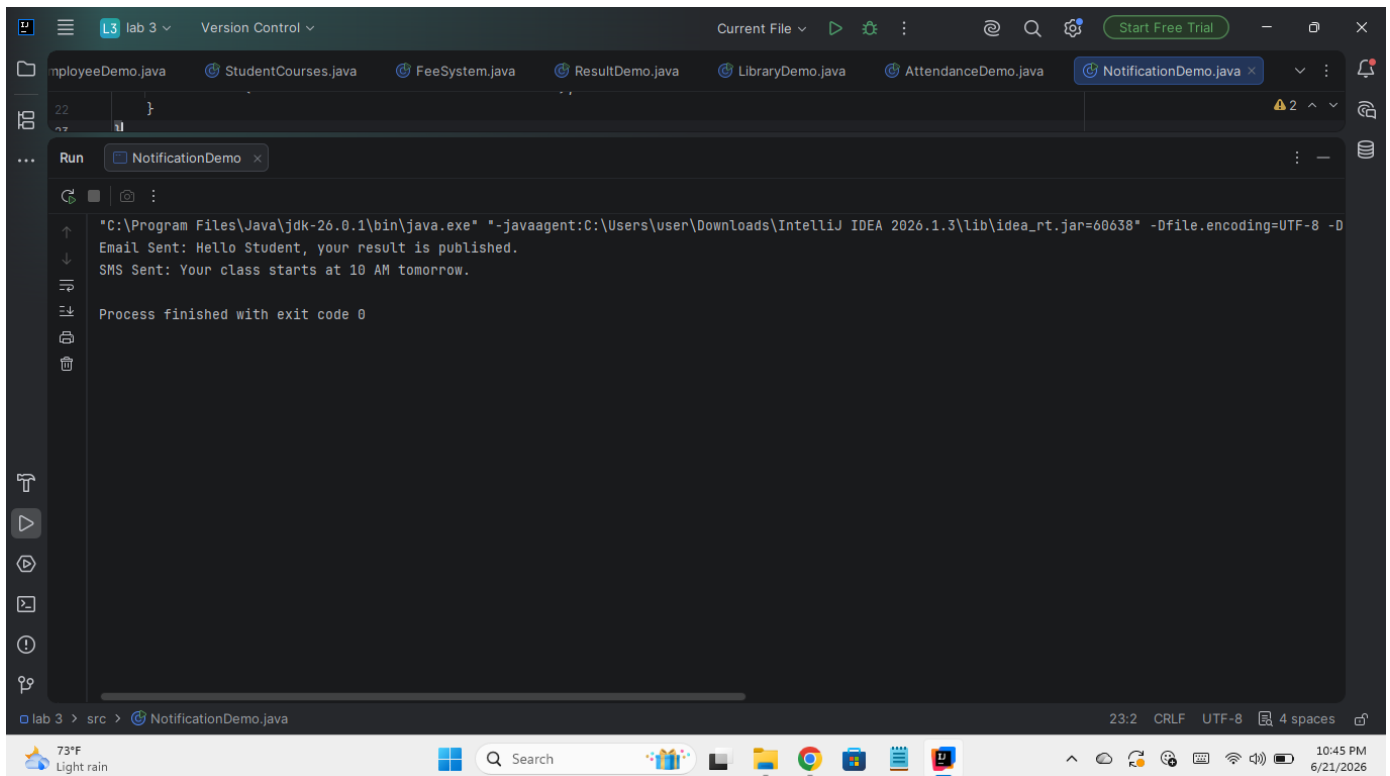


```
1 interface Notification { 4 usages 2 implementations
2     void send(String message); 2 usages 2 implementations
3 }
4 class EmailNotification implements Notification { 1 usage
5     @Override 2 usages
6     public void send(String message) {
7         System.out.println("Email Sent: " + message);
8     }
9 }
10 class SMSNotification implements Notification { 1 usage
11     @Override 2 usages
12     public void send(String message) {
13         System.out.println("SMS Sent: " + message);
14     }
15 }
16 public class NotificationDemo {
17     public static void main(String[] args) {
18         Notification email = new EmailNotification();
19         Notification sms = new SMSNotification();
20         email.send("Hello Student, your result is published.");
21         sms.send("Your class starts at 10 AM tomorrow.");
22     }
23 }
24 }
```

Run NotificationDemo

lab 3 > src > NotificationDemo.java 23:2 CRLF UTF-8 4 spaces

Output:



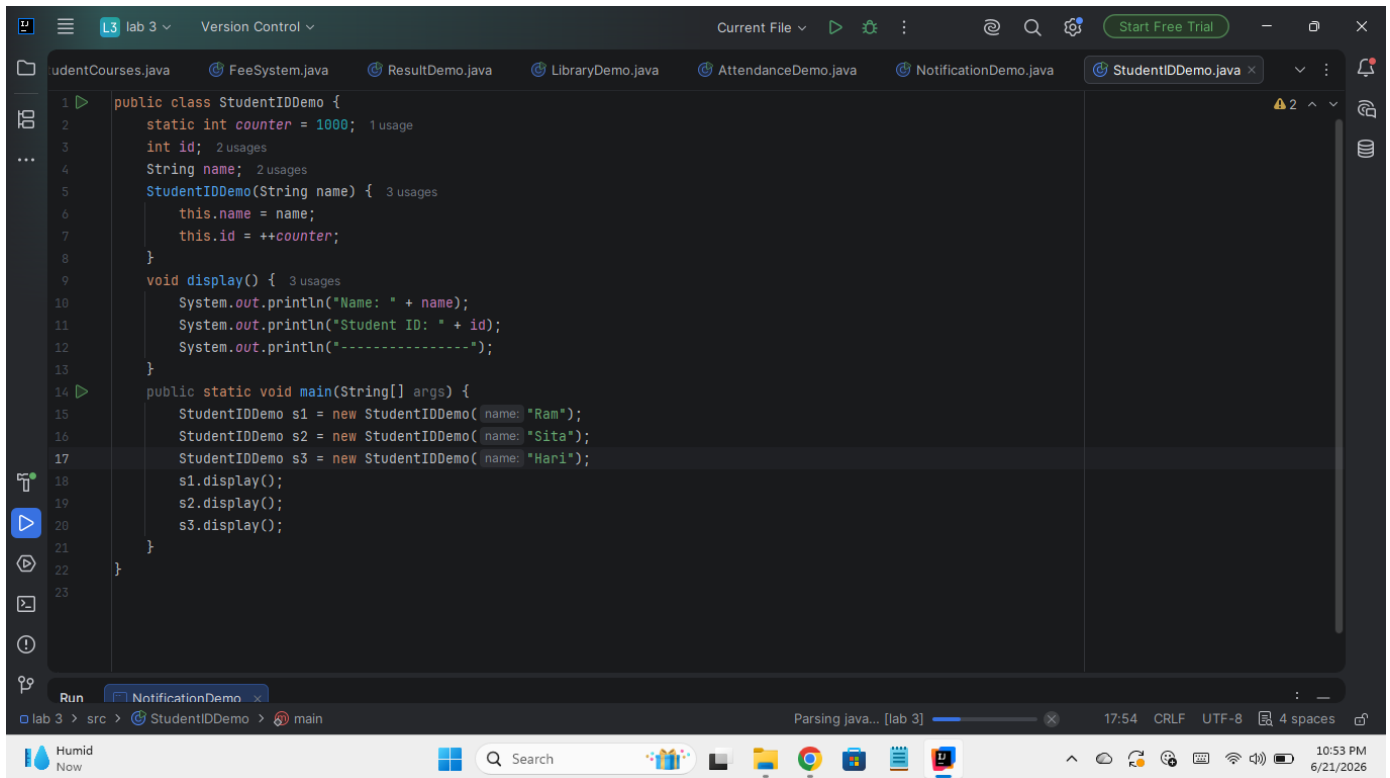
```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\Downloads\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=60638" -Dfile.encoding=UTF-8 -D
Email Sent: Hello Student, your result is published.
SMS Sent: Your class starts at 10 AM tomorrow.

Process finished with exit code 0
```

lab 3 > src > NotificationDemo.java 23:2 CRLF UTF-8 4 spaces

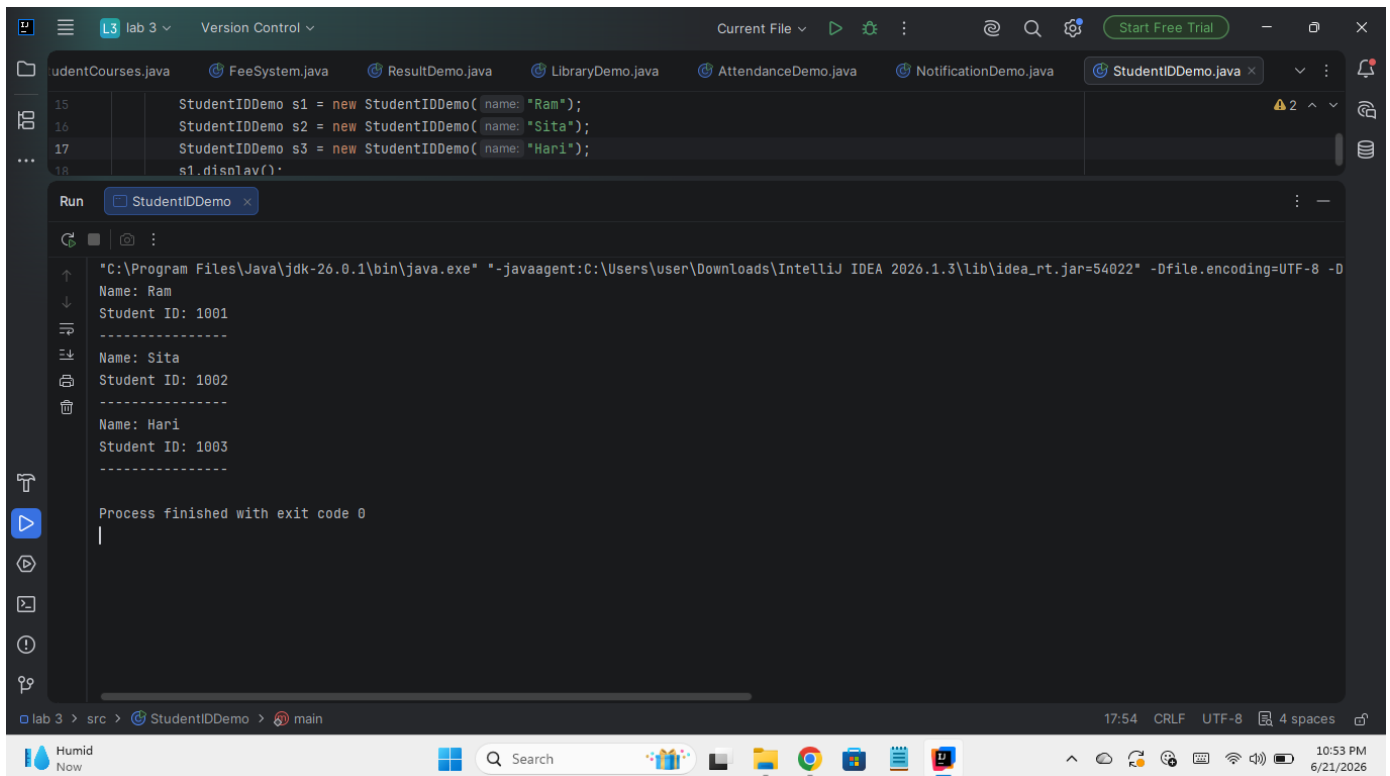
9 Unique ID

Program:



```
1 public class StudentIDDemo {
2     static int counter = 1000;
3     int id;
4     String name;
5     StudentIDDemo(String name) {
6         this.name = name;
7         this.id = ++counter;
8     }
9     void display() {
10        System.out.println("Name: " + name);
11        System.out.println("Student ID: " + id);
12        System.out.println("-----");
13    }
14    public static void main(String[] args) {
15        StudentIDDemo s1 = new StudentIDDemo("Ram");
16        StudentIDDemo s2 = new StudentIDDemo("Sita");
17        StudentIDDemo s3 = new StudentIDDemo("Hari");
18        s1.display();
19        s2.display();
20        s3.display();
21    }
22 }
23
```

Output:



```
15 StudentIDDemo s1 = new StudentIDDemo("Ram");
16 StudentIDDemo s2 = new StudentIDDemo("Sita");
17 StudentIDDemo s3 = new StudentIDDemo("Hari");
18 s1.display();

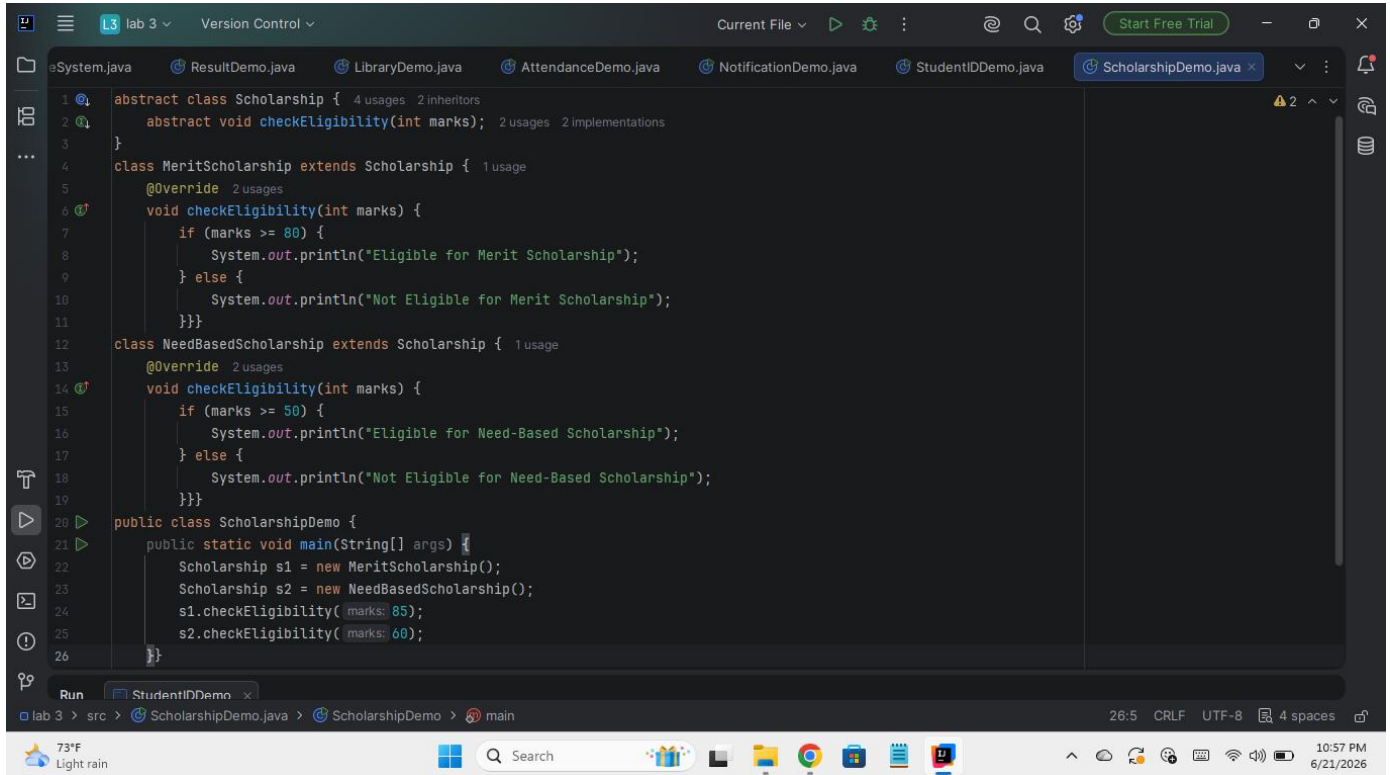
Run StudentIDDemo

"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\Downloads\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=54022" -Dfile.encoding=UTF-8 -D
Name: Ram
Student ID: 1001
-----
Name: Sita
Student ID: 1002
-----
Name: Hari
Student ID: 1003
-----

Process finished with exit code 0
```

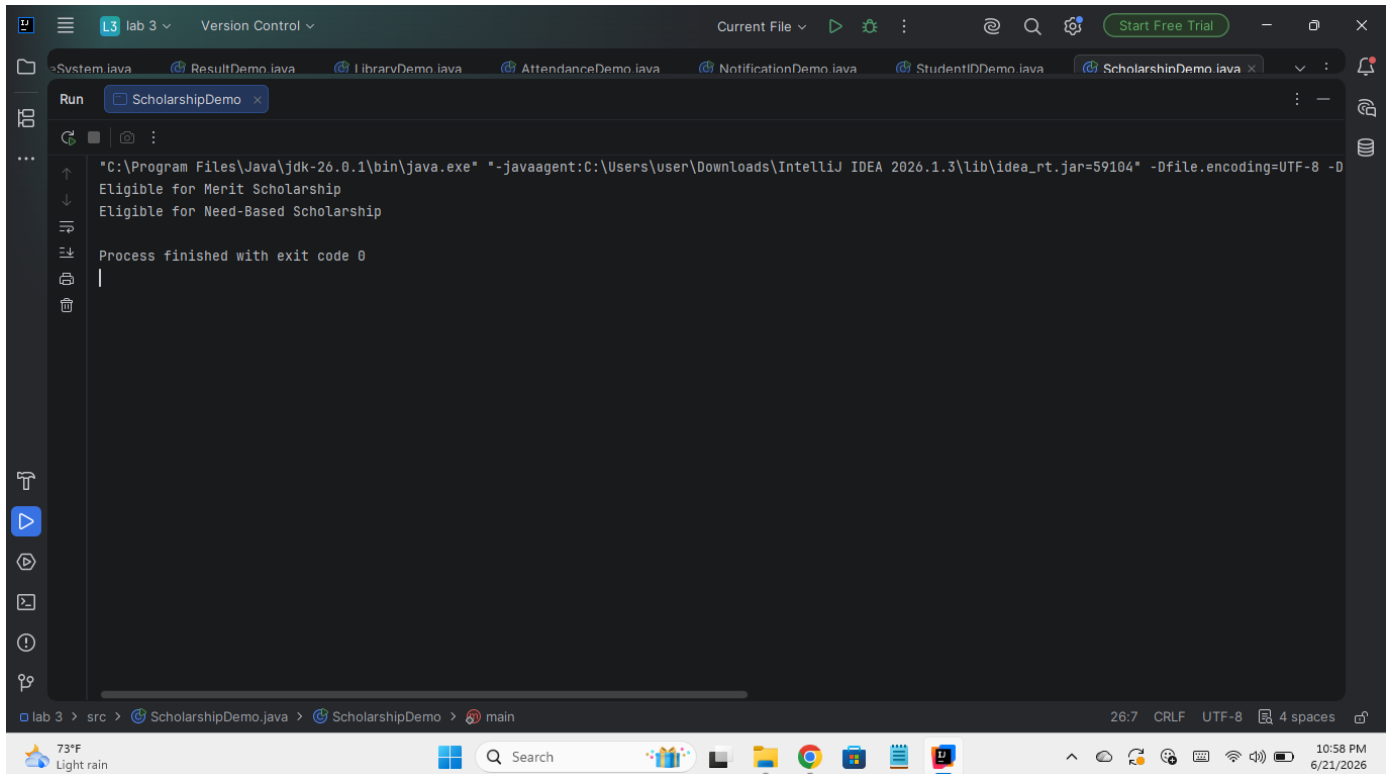
10 Scholarship eligibility

Program:



```
1 abstract class Scholarship { 4 usages 2 inheritors
2     abstract void checkEligibility(int marks); 2 usages 2 implementations
3 }
4 class MeritScholarship extends Scholarship { 1 usage
5     @Override 2 usages
6     void checkEligibility(int marks) {
7         if (marks >= 80) {
8             System.out.println("Eligible for Merit Scholarship");
9         } else {
10            System.out.println("Not Eligible for Merit Scholarship");
11        }
12    }
13 }
14 class NeedBasedScholarship extends Scholarship { 1 usage
15     @Override 2 usages
16     void checkEligibility(int marks) {
17         if (marks >= 50) {
18             System.out.println("Eligible for Need-Based Scholarship");
19         } else {
20             System.out.println("Not Eligible for Need-Based Scholarship");
21        }
22    }
23 }
24 public class ScholarshipDemo {
25     public static void main(String[] args) {
26         Scholarship s1 = new MeritScholarship();
27         Scholarship s2 = new NeedBasedScholarship();
28         s1.checkEligibility(marks: 85);
29         s2.checkEligibility(marks: 60);
30     }
31 }
```

Output:



```
Run ScholarshipDemo x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\Downloads\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=59104" -Dfile.encoding=UTF-8 -D
Eligible for Merit Scholarship
Eligible for Need-Based Scholarship
Process finished with exit code 0
```